

Reinventing asset servicing with distributed ledger technology



HSBC

| Opening up a world of opportunity

Reinventing asset servicing with distributed ledger technology

Received (in revised form): 6th September, 2023

Abstract

This paper draws on the hands-on experience of HSBC in implementing distributed ledger technology (DLT)-based solutions through building its own proprietary tokenisation platform, HSBC Orion, and participating in various industry initiatives such as the Hong Kong Monetary Authority (HKMA)'s inaugural tokenised green bond issuance. The paper highlights key considerations for building DLT-based solutions and provides a view of the future state of digital assets as the ecosystem continues to grow and evolve.

Keywords: distributed ledger technology, tokenisation, digital assets

Journal of Securities Operations & Custody Vol. 16, No. 2, pp.146–154
© Henry Stewart Publications,
1753-1802

HSBC, 10 Marina Boulevard,
Marina Bay Financial Centre Tower 2,
#45-01, Singapore 018983





Introduction

Asset servicing as an industry is at a unique inflection point. In the past two decades, the industry has largely progressed on incremental innovation by improving efficiency, automating existing services and shortening settlement cycles; however, it is now on the cusp of major reinvention. The catalyst for this change is innovation in the way assets can be represented. Tokenisation of traditional assets can be seen as the next evolution in asset representation. The tokenisation of assets involves ‘the digital representation of real (physical) assets on distributed ledgers, or the issuance of traditional asset classes in tokenised form’.¹

Just as the shift from physical to dematerialised forms of asset representation changed capital markets, the adoption of tokenised assets will improve efficiencies, accelerate growth and encourage product innovation (see Figure 1). Almost any asset in the world can be tokenised and represented on distributed ledgers, allowing these assets to be fractionalised and transferred across different asset networks. Examples include traditional assets such as debt and equity (both listed and private) and alternative assets such as carbon credits, precious metals, collectibles and artwork.

In recent years, traditional financial institutions have been making significant investments into distributed ledger technology (DLT)-based infrastructure. The value of the technology has been proven through a number of successful

experiments in the industry. The focus has now shifted from proof of concepts to commercialisation and development of production-grade systems and infrastructure.

In 2023, HSBC launched HSBC Orion, HSBC’s proprietary tokenisation platform for issuance and servicing of digital assets. On 31st January, 2023, the European Investment Bank (EIB) issued the pilot tokenised bond on HSBC Orion, under Luxembourg law.² Apart from launching this platform, the bank also participated in other asset tokenisation networks. HSBC collaborated with the Hong Kong Monetary Authority (HKMA)³ on their tokenised green bond issuance for the Government of the Hong Kong Special Administrative Region of the People’s Republic of China (HKSAR) on Goldman Sachs’ Digital Asset Platform, GS DAP™. HSBC was one of the participating banks and played a few roles, including market practice adviser, joint lead manager, lodging and paying agent and custodian.

In Singapore, HSBC is participating in Monetary Authority of Singapore (MAS) Project Guardian,⁴ a collaborative initiative with the financial industry to explore the economic potential, value of asset tokenisation and decentralised finance (DeFi).

Participation in these initiatives has informed the general understanding of the key design considerations and principles that have to be taken into account in the development of blockchain-based asset networks.



Figure 1 Advantages of tokenisation

Key design considerations and principles

Technology

One of the first decisions to make when building a new asset network is the choice of the underlying blockchain. Blockchain selection is listed as a key early decision for securities tokenisation by Asia Securities Industry & Financial Markets Association (ASIFMA), with relevant factors to consider including security, speed of transfer and accessibility.⁵ While public permissionless blockchains such as Ethereum offer transparency and visibility, the lack of restrictions for participation may pose know-your-customer (KYC)/anti-money laundering (AML) concerns. On the other hand, private permissioned blockchains allow for access and participants to be controlled but offer less transparency. HSBC Orion is operated from Luxembourg, and under the Luxembourg legal framework, one of the responsibilities that HSBC has in its role as registrar or

‘central account keeper’ on the platform is ensuring that participants on the platform meet KYC/AML standards.

To address the dichotomy between the objective of transparency and the need to govern the accession and termination of parties in the network, HSBC eventually decided on using two blockchains: Hyperledger Fabric (private) and Ethereum (public) (see Figure 2).

The private ledger serves as the legally definitive record of title to the digital bonds (and settlement tokens) issued on HSBC Orion, and the public side serves as an information source to the issuer and investors. Participants would be able to view the history of transactions for the digital bond, as well as holdings of digital bonds corresponding to their own wallets on Ethereum.

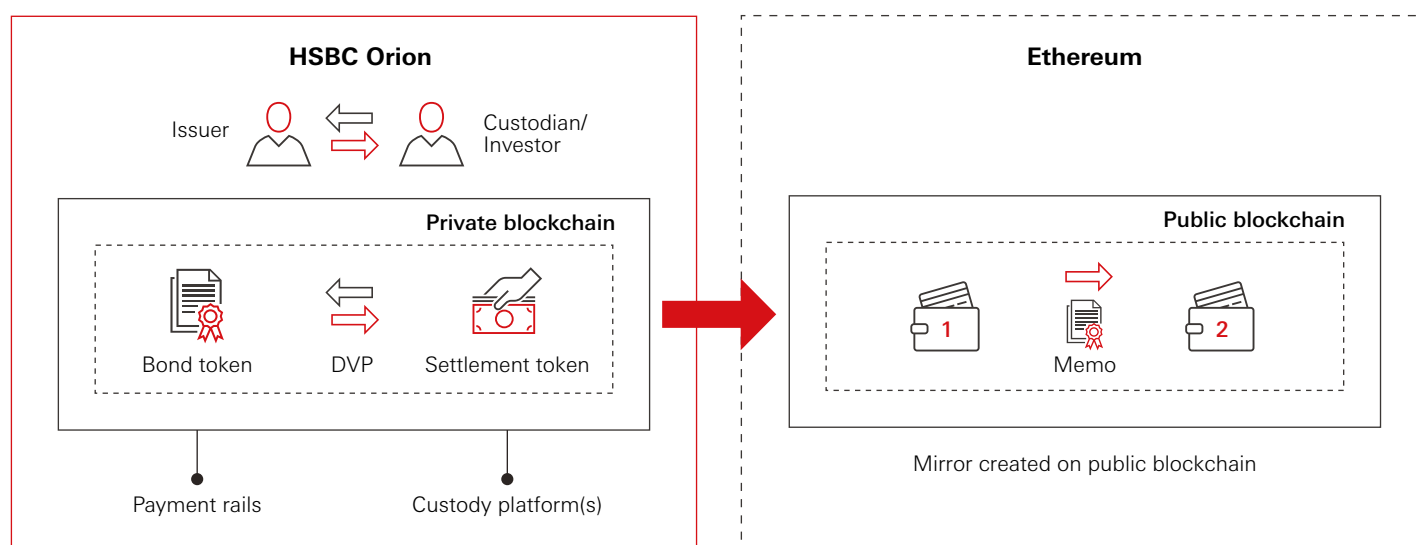


Figure 2 HSBC Blockchains

Token design is another integral part of the asset network; four tokens were designed for HSBC Orion (see Figure 3).

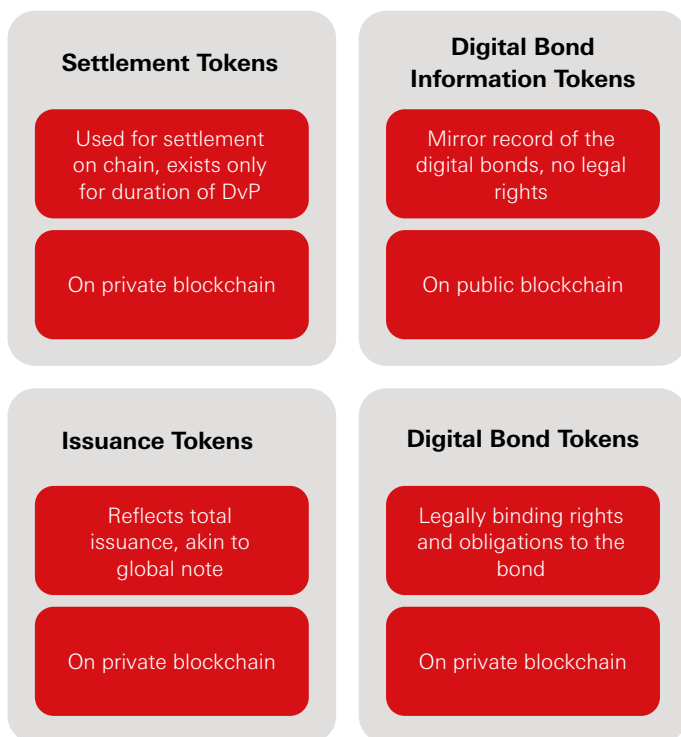


Figure 3 HSBC Orion tokens

The first three tokens exist on the private blockchain and support the creation of legally binding rights and obligations on the private blockchain, whereas the last token is present on the public blockchain and is for informational purposes only. The issuance token reflects the native minting of a financial instrument and records the identification elements of a digital bond issuance on the platform — akin to the registration of a global note with a common depository in the traditional process. The digital bond tokens are classified as digital-native securities. The settlement tokens are ephemeral, used and exist for the period of delivery versus payment (DvP) on-chain, and are records of fiat cash moved for settlement purposes.

Lastly, the digital bond information tokens on Ethereum serve as a mirror record of the digital bonds, providing transparency to investors. These tokens have no legal rights attached and have been designed in accordance with the ERC-20 standard.

Digital assets modelling language (DAML), an open-source smart contract language, was used, making future integrations possible through enabling interoperability across enterprise blockchains.

HSBC Orion was also designed for distributed nodes. This gives the participants on HSBC Orion the ability to host their own nodes, which will allow for greater privacy and security and more configurability (eg ability to write their own smart contracts automating some of their interactions with platform activity). This is also aligned to HSBC's vision of building for interoperability between asset networks.

Operations

One of the key objectives of developing a tokenised asset network was to improve operational efficiencies, particularly at the clearing and settlement stage of the asset life cycle. While progress has been made in recent years to shorten settlement cycles, there is a limit to what can be achieved with the current legacy systems and infrastructure.

A key goal for HSBC Orion was to achieve 'atomic' or instant settlement between cash and asset tokens, which was accomplished by the use of smart contracts. On HSBC Orion, smart contracts trigger DvP transactions between cash and digital bonds. The smart contract checks that the relevant participant has sufficient funds in its fiat cash account held with HSBC. If so, the system creates a corresponding cash settlement token. The cash settlement token and bond tokens were both moved to intermediary accounts before being transferred to their destination accounts, following where the cash token was removed from the system and reflected back as fiat cash. This enabled instant gross settlement as long as sufficient fiat cash and digital bond tokens were present in the respective digital accounts. This model removes the need for batch settlement cycles, reduces settlement risk, and has resulted in near atomic settlement without the need to use a digital currency on-chain (see Figure 4).

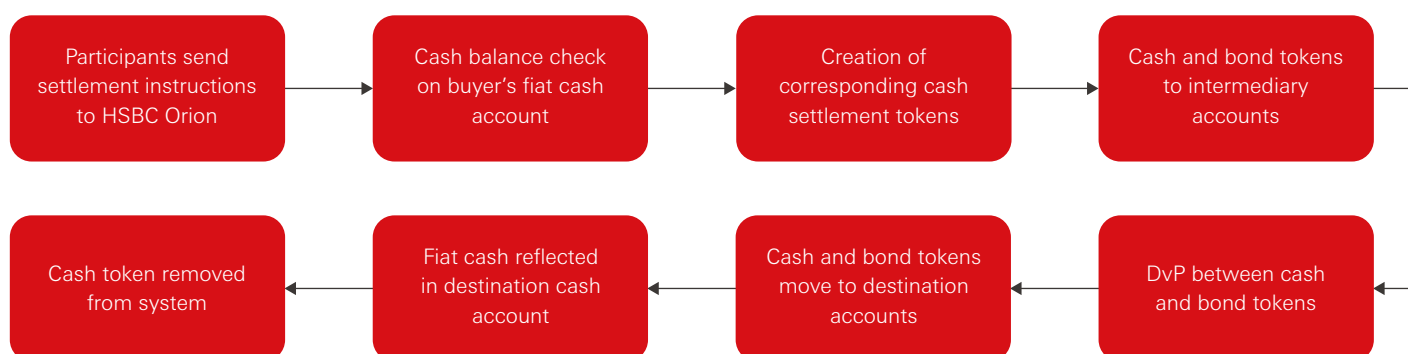


Figure 4 DvP settlement on HSBC Orion

Despite the progress made, there is still room for further growth. Currently, custodians are directly connected to HSBC Orion but end-investors are not. While direct connectivity of all market participants to tokenised asset networks would be ideal, this is often challenging in practice. Many issuers and investors do not currently have the resources or the inclination to host their own node on a blockchain network. This limits the extent of automation that is realistically possible.

In fact, one of the key challenges operationally was interfacing with the existing legacy systems currently being used by market participants through the end-to-end asset life cycle. The Bank for International Settlements (BIS) highlights in a recent paper the technical challenges in terms of the design of 'ramps' connecting traditional systems with tokenisation platforms, explaining that the involvement of offline intermediaries to achieve full automation is challenging.⁶ On HSBC Orion, it would not have been feasible to move all operational processes on-chain, so data is still needed to be fed into and from HSBC Orion to existing systems at different stages of the issuance and settlement process.

Development of new asset networks needs to be scoped to ensure that DLT is being applied where its full benefits can be realised. For the digital bonds issued on HSBC Orion, since the main focus was on achieving atomic settlement, a fully digitised book building and pricing process was kept off-platform and was carried out by relying on existing systems and processes. Building an integration layer to connect to existing systems using current channels such as SWIFT will be a key focus, as HSBC Orion will continue to be scaled. Connectivity to existing infrastructure is likely to remain essential for blockchain-based asset networks, since both traditional and digital assets are likely to exist together in a hybrid ecosystem for some time.

Regulatory and governance

To create trust and drive adoption by issuers, investors and financial institutions, it is critical for blockchain-based asset networks to operate under a clearly defined legal and regulatory framework. A fully digital native asset network operates under a legal framework which recognises tokenised securities as having the same legal status as traditional securities. HSBC Orion's legal framework is tailored to allow for the issuance, transfer and custody of securities on DLT infrastructure. The Luxembourg law

on Dematerialised Securities 2013, as amended by the Blockchain Act, is centred around a central account keeper and allows a distributed ledger to be used as the legal record of title to the assets, removing the need for a physical global note to be issued to represent the bond, thus supporting a fully native digital bond issuance on a DLT platform. A thorough legal and regulatory review was conducted to ensure that the DLT platform functionalities and the roles and responsibilities of the participants complied with the regulatory framework.

Besides regulation, compliance and monitoring requirements need to be considered. A key component of this is authentication and permissioning. On HSBC Orion, the use of the private ledger allows for a greater degree of control in terms of participant onboarding onto the network. Custodians and arrangers go through a standard KYC and due diligence process before being granted access to the network. Defined user roles with specific permissions are created to ensure that users have appropriate access to data on the platform. All transactions have to be initiated from the private ledger before being mirrored to the public ledger and approval rights to authorise recording of transactions to Ethereum is tightly controlled.

Another consideration is transaction monitoring especially in relation to financial crime considerations. For tokenised versions of traditional securities that have an identifier, eg ISIN, existing compliance tools can be used to screen the security itself. In addition, monitoring may be needed specifically on wallets and transactions on public blockchains. For transactions recorded on HSBC Orion, the risk is mitigated as transactions could take place only between wallets linked to participants onboarded to the private network.

To record transactions on HSBC Orion onto Ethereum, gas fees have to be paid to miners on the Ethereum network. On public blockchains, it is not possible to conduct a screening on miners' wallet addresses before mining is completed. This is one area where additional clarity in regulatory guidance on the obligations of asset network operators could be beneficial. The compliance and monitoring space for blockchain networks is evolving quickly and there is a range of powerful blockchain analytic tools and services emerging to monitor, detect and respond to fraud, money laundering, sanctions evasion and other risks.

Building the future of digital assets



While the industry has made progress, most innovations have been made within walled gardens. A number of financial institutions have developed their own tokenisation platforms but as of today there is limited connectivity between them. In order to accelerate growth in this space, we believe interoperability, scalability and legal clarity will be key building blocks of the future of digital assets.

Legal and regulatory clarity

There has been significant advancement in legal and regulatory clarity for blockchain and digital assets in recent years; however, regulatory change is fragmented across jurisdictions. For example, different regulators have different token classification frameworks and differing standards on permissibility of digital native tokens. Regulatory alignment and harmonisation across jurisdictions is critical to provide the clarity needed for global blockchain asset networks to thrive.

Interoperability

Interoperability and collaboration will be crucial to drive greater institutional adoption of DLT applications. Enabling interoperability has been identified by the Global Financial Markets Association (GFMA) as one of the imperatives to achieving network effects for tokenised asset networks. The paper elaborates that a framework of standards is needed to guide market-wide interoperability.⁷ There are multiple dimensions in realising interoperability.

This could include interoperability by using the same technical stack (ie same smart contract language, base technical infrastructure, etc.) or building integration layers between networks on different underlying blockchains. One way to enable this is standardisation of token formats across networks. The ERC-20 (fungible tokens), ERC-721 (non-fungible tokens) and ERC-1155 (multi-token) standards are widely recognised in the ecosystem and enable the tokens to be used across different Ethereum Virtual Machine (EVM) compatible blockchains.

Achieving scale

One of the hurdles to achieving wider market adoption of blockchain-based asset networks is the lack of liquidity of the assets issued on these platforms. Most legal frameworks currently do not allow the admission of digital assets to a recognised trading venue, and therefore most of the digital assets natively created on a DLT can be traded over the counter (OTC) only between participants connected to the network. Establishing connectivity with exchanges and liquidity venues is essential to improve adoption among a broader range of issuers and investors. Investors should be able to place trades on existing trading venues and have the settlement instruction flow straight through to tokenisation platforms such as HSBC Orion for settlement. In addition to trading venues, connectivity to other applications, for example collateral and repurchase agreement (repo) networks, can improve the utility of the assets, incentivising adoption.



Recommendations

We have listed below several recommendations for companies to consider when designing and developing DLT-enabled platforms, based on our experience of building a production-ready tokenisation platform within a regulated financial services context.

Clear scope and objectives

One of the most important factors for success is defining clear objectives and scope for the platform. This would inform design considerations such as which functions would be performed 'on-chain', what data needs to be reflected on the blockchain, and which functions can be done off-platform with relevant data being fed onto the platform subsequently. For Orion, since the key aim was the minting (issuance of bond tokens) and the achievement of atomic settlement; this informed decision making on the processes that should be automated via smart contracts and the data that should be recorded on the private and public blockchains.

Design for the future

The digital assets and tokenisation space is still at an early stage of development, and it is difficult to envision what the best-in-class technology and operating models will look like five or even two years into the future. Hence it is especially important that DLT-enabled platforms are developed following a modular approach, with interconnected components that can be swapped out and replaced in future without massive development effort. The ultimate goal for blockchain-based platforms is often to create a truly distributed, decentralised network with each network participant able to control their own data and write their own smart contracts and business logic on the platform. Companies should consider their

overall platform strategy to ensure that even though in the short run, distribution of nodes is not practicable for participants, the platform is designed with an approach that enables decentralisation in the medium to long term.

Defining network structure

A key point to consider when designing new digital asset networks is network structure and its impact on interoperability with other networks and the ecosystem as a whole. In their paper on Open and Interoperable Networks,⁸ the Monetary Authority of Singapore provides a useful framework on network structure and interactions. The paper outlines three models of network interactions: flat networks, where participants interact with a common ledger; layered networks, where participants process transactions separately on 'upper layers' while a summary is posted to a common 'lower layer'; and interlinked networks, which refer to a network of independent networks or sidechains. These models have varying pros and cons when it comes to transaction speed, ease of adoption and extent of decentralisation.

HSBC Orion in its current implementation contains elements of the flat network model where direct participants access the common private ledger. As we continue to develop the platform, we are considering the evolution of network interactions and the path toward a future state where HSBC Orion is integrated with multiple digital asset networks. Companies should consider not just their immediate objectives but also their future vision for their digital assets network when deciding how to structure their network and facilitate network interactions between participants.

The evolution of digital assets: what is ahead?

DLT by its very nature requires collaboration across multiple participants in order to realise the full benefits of the network. We see a future of multiple interconnected asset networks where tokens can be moved and used freely across the ecosystem. In the last decade or so, the industry has mostly focused on incremental innovation and efficiency gains — essentially building a faster horse. It is time to shift gears from carriages to engines, and to build completely new business and operating models.



Authors



Zhu Kuang Lee

Managing Director – Chief Digital, Data and Innovation Officer, Securities Services, HSBC, Singapore
Tel: +65 6658 7435
E-mail: zhu.kuang.lee@hsbc.com.sg



Jing Yu Wong

Senior Product Manager, Digital, Data and Innovation, Securities Services, HSBC, Singapore
Tel: +65 6658 0104
E-mail: jing.y.wong@hsbc.com.sg



Rachel Roch

Senior Product Manager, Digital, Data and Innovation, Securities Services, HSBC, Singapore
Tel: +65 6658 0174
E-mail: rachel.roch@hsbc.com.sg



Xin Yi Tan

Senior Product Manager, Digital, Data and Innovation, Securities Services, HSBC, Singapore
Tel: +65 6658 1983
E-mail: xin.yi.tan@hsbc.com.sg

Zhu Kuang Lee is the Chief Digital, Data and Innovation Officer for HSBC's Securities Services, where he is responsible for defining and driving the digital, data, innovation, platform and partnership strategies and digital product vision globally.

Jing Yu Wong is passionate in showcasing HSBC's thought leadership and views on emerging technologies that help their clients embrace change and create value. Together with the Digital, Data and Innovation team, she creates thought leadership about emerging technology trends such as blockchain and digital assets that have the potential to shape the future of financial services.

Rachel Roch has driven product design and development for a number of global, crossindustry projects in the digital assets and tokenisation space, including HSBC Orion. As part of the digital and data team, Rachel helps shape the digital agenda within the business, partnering with clients to leverage the full potential of emerging technologies, with a particular focus on blockchain and digital assets.

Xin Yi Tan, together with the Digital, Data and Innovation team, has been involved in ground-breaking work in the distributed ledger technology (DLT) space and driving the digital agenda for the business. Her area of focus is developing digital solutions leveraging HSBC's banking expertise to meet client demands in the growing wealth industry as well as digital assets ecosystem globally.

Acknowledgments

We would like to acknowledge the core cross-functional teams that worked to make the HSBC Orion platform a reality.

Securities Services, Digital and Data: Zhu Kuang Lee, Rajeev Tummala, Maria Cabanting, Hashir Rahim, Rachel Roch and Xin Yi Tan

Digital Assets Strategy: John O'Neill

Technology: Chris Jones, Sandeep Kothale, Andrew Hon, Mark Heaton

Operations: Stuart Warner, Christophe Decker and Gerard Kayser

Debt Capital Markets: Asif Sherani

Product Delivery: Michael Yuen

Legal: Michael Lehotzki

References

- 1 The Organisation for Economic Co-operation and Development (January 2020), 'The Tokenisation of Assets and Potential Implications for Financial Markets', available at <https://www.oecd.org/finance/The-Tokenisation-of-Assets-and-Potential-Implications-for-Financial-Markets-HIGHLIGHTS.pdf> (accessed 16th May, 2023).
- 2 European Investment Bank (EIB) (January 2023), 'EIB issues its first ever digital bond in pound sterling', available at <https://www.eib.org/en/press/all/2023-030-eib-issues-its-first-ever-digital-bond-in-british-pounds> (accessed 16th May, 2023)
- 3 Hong Kong Monetary Authority (HKMA) (February 2023), 'HKSAR Government's inaugural tokenised green bond offering', available at <https://www.hkma.gov.hk/eng/news-and-media/press-releases/2023/02/20230216-3/> (accessed 16th May, 2023).
- 4 Hong Kong Monetary Authority (HKMA) (November 2022), 'First Industry Pilot for Digital Asset and Decentralised Finance Goes Live', available at <https://www.mas.gov.sg/news/media-releases/2022/first-industry-pilot-for-digital-asset-and-decentralised-finance-goes-live> (accessed 26th May, 2023).
- 5 Asia Securities Industry & Financial Markets Association (ASIFMA) (November 2019), 'Tokenised Securities, A Roadmap for Market Participants and Regulators', available at <https://www.asifma.org/wp-content/uploads/2019/11/tokenised-securities-a-roadmap-for-market-participants-final.pdf> (accessed 26th May, 2023)
- 6 Bank for International Settlements (BIS) (April 2023), 'The tokenization continuum', BIS Bulletin, available at <https://www.bis.org/publ/bisbull72.pdf> (accessed 26th May, 2023).
- 7 Global Financial Markets Association (May GFMA) (2023), 'Impact of Distributed Ledger Technology in Global Capital Markets', available at <https://www.gfma.org/policies-resources/gfma-publishes-report-on-impact-of-dlt-in-global-capital-markets/> (accessed 26th May, 2023).
- 8 Monetary Authority of Singapore (June 2023), 'Project Guardian, Enabling Open and Interoperable Networks', available at <https://www.mas.gov.sg/-/media/mas-media-library/development/fintech/project-guardian/project-guardian-open-interoperable-network.pdf> (accessed 26th May, 2023).

